A Report Based On

"ON JOB TRAINING"

Carried out at

Bajaj Allianz Life Insurance, Pune

Entitled

"A Rule-Based Insurance Policy Recommendation System Using Apriori Algorithm"

Submitted at

Department of Statistics
School of Mathematical Sciences
Kavayitri Bahinabai Chaudhari North Maharashtra University, Jalgaon

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Abstract

In the dynamic domain of life insurance, selecting the most appropriate policy configuration—especially the benefit term—remains a challenge for both customers and agents. This project presents a dual model insurance recommendation system designed to improve transparency and precision in policy suggestions. The system leverages both a rule-based engine built on domain expertise and structured logic, and a data-driven model utilizing the Apriori algorithm for association rule mining via the **MLxtend** library in Python.

To simulate real-world scenarios, synthetic datasets were generated based on demographic and policy trends. The rule-based model applies predefined decision tables, while the Apriori model dynamically uncovers patterns between premium terms and benefit terms across diverse customer profiles. The recommendation engine is deployed using both **Streamlit** (Python) and **R Shiny**, offering flexible and interactive user interfaces for analysts and business users alike.

This hybrid framework demonstrates the potential to combine human insight, data mining, and modern web technologies to support intelligent and explainable policy recommendations in the insurance sector.

Acknowledgment

We would like to take this opportunity to express our sincere gratitude to all those who contributed to the successful completion of our internship at **Bajaj Allianz Life Insurance, Pune**, particularly in the **Actuarial Department**.

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Furthermore, we would also like to extend our sincere thanks to **Prof. Manoj C. Patil**, our mentor from the Department of Statistics of the School of Mathematical Sciences, Kavayitri Bahinabai Chaudhari North Maharashtra University, Jalgaon, for his academic mentorship, support, and guidance throughout the month.

We would also like to thank our teammates and fellow classmates who participated in the OJT and project at Bajaj Allianz for their cooperation, ideas, and teamwork, which made the entire experience more enriching and collaborative.

This internship has been an enriching experience, and we are grateful to everyone who participated in it.

Yours Sincerely,
Miss. Rajashri Durvas Patil
Miss. Trupti Bhatu Patil
Mr. Aditya Kishorsing Pardeshi

Chapter 1

Comprehensive Analysis of Bajaj Allianz Life Insurance: Data Insights and Organizational Overview

1.1 Introduction

This report presents a comprehensive analysis of life insurance product data for **Bajaj Allianz Life Insurance**, conducted as part of an On-the-Job Training (OJT) assignment. The project integrates data-driven techniques with an in-depth understanding of the organization's operations, product portfolio, and strategic initiatives. The analysis is structured into two primary components:

- **Data Analysis and Exploration:** Generation and examination of synthetic life insurance data reflective of Bajaj Allianz's product mix, employing *Exploratory Data Analysis (EDA)* to identify patterns in customer behavior, product preferences, and risk indicators across demographic and financial segments.
- **Recommendation Model Development:** Implementation of a rule-based recommendation system using the **Apriori algorithm** to propose tailored insurance product bundles, enhancing personalization and optimizing product targeting.

Additionally, this report provides insights into Bajaj Allianz Life's organizational structure, digital transformation efforts, distribution strategies, and key performance metrics, such as claim settlement efficiency. The project aims to bridge academic knowledge with practical applications, delivering actionable insights to support strategic decision-making in insurance product design and customer engagement.

1.2 Objectives and Purpose of the Internship

The internship was designed to align theoretical knowledge with industry practices, fostering professional development and industry-specific expertise. The objectives and purposes are outlined as follows:

- **Bridging Academic Knowledge with Industry Practice:**
  - *Objective:* Apply statistical and analytical techniques to address real-world challenges in the life insurance sector.

- *Purpose:* Gain practical experience in customer segmentation, policy analysis, and data-driven decision-making within a business context.

- **Gaining Industry-Specific Exposure:**
  - *Objective:* Understand the operational structure, workflows, and functional areas of a leading life insurance company.
  - *Purpose:* Develop familiarity with critical insurance processes, including premium collection, risk assessment, and policy servicing, through cross-departmental collaboration.

- **Applying Analytical Tools and Techniques:**
  - *Objective:* Utilize advanced analytical tools and algorithms to analyze business data.
  - *Purpose:* Build proficiency in Python and Excel through hands-on application in projects such as the Apriori-based recommendation system.

- **Professional Development and Workplace Readiness:**
  - *Objective:* Enhance soft skills and corporate professionalism.
  - *Purpose:* Improve communication, time management, and teamwork capabilities through mentorship and collaborative projects.

- **Contributing to Organizational Goals:**
  - *Objective:* Participate in impactful projects that support business outcomes and explore career opportunities in analytics, actuarial science, and insurance.
  - *Purpose:* Deliver value through a recommendation engine that enhances customer engagement and personalization, aligning academic training with industry needs to prepare for future roles.

1.3 Overview of Bajaj Allianz Life Insurance

1.3.1 Company Profile

Bajaj Allianz Life Insurance Company Limited, established in August 2001 and headquartered in Pune, Maharashtra, is a leading private life insurer in India. A joint venture between **Bajaj Finserv Ltd.**, a prominent Indian non-banking financial company, and **Allianz SE**, a global leader in insurance and asset management based in Germany, the company combines global expertise with localized market insights to deliver innovative, customer-centric life insurance solutions.

1.3.2 Mission and Vision

- **Mission:** To provide value-driven, customer-focused life insurance solutions that ensure the financial security and well-being of Indian families.
- *Vision:* To be a transparent, trustworthy, and digitally advanced insurer that consistently exceeds customer expectations.

1.3.3 Product Portfolio

Bajaj Allianz Life offers a diverse range of products tailored to various life stages and financial goals, including:

- **Term Insurance Plans:** Policies such as *Smart Protect Goal*, offering benefits like critical illness coverage, accidental death benefits, return of premium, and waiver of premium on disability.
- **Unit Linked Insurance Plans (ULIPs):** Investment-linked plans like *Goal Assure* (featuring loyalty additions, return of mortality charges, multiple fund choices, and return enhancer on maturity) and *Future Gain* for balanced growth and protection.
- **Savings and Endowment Plans:** Traditional plans such as *Save Assure* (guaranteed maturity payout) and *Guaranteed Income Goal* (regular income post-maturity).
- **Child Plans:** Policies like *Young Assure*, providing milestone payouts and waiver of premiums in case of the parent's death.
- **Retirement and Pension Plans:** Plans such as *Retire Rich*, offering flexible annuity options, including life annuity, annuity with purchase price return, and joint life options for spouses.
- **Health Riders:** Supplementary benefits, including critical illness, waiver of premium, and accidental death benefit riders.

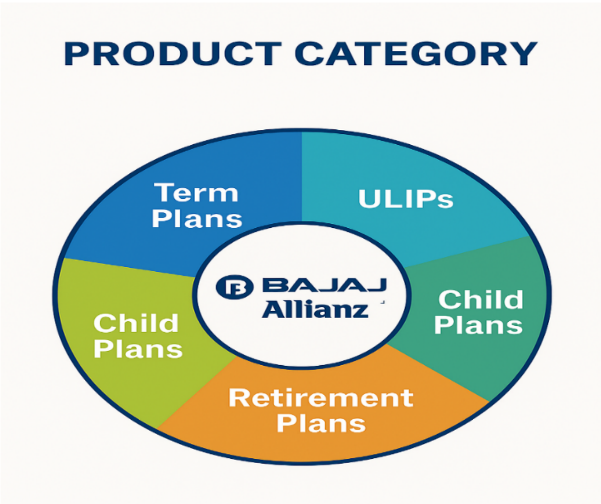


Figure 1.1: Distribution of Bajaj Allianz Life Insurance product categories

1.3.4 Digital Innovation and Services

Bajaj Allianz Life is at the forefront of digital transformation, offering:

- Online policy management and premium payment portals.
- Mobile app and WhatsApp-based customer servicing.
- AI-powered support for enhanced customer experience.

1.3.5 Awards and Recognition

The company has earned numerous accolades for its excellence in service and innovation, including:

- **Customer Service Excellence:** Recognized for seamless and efficient customer experiences across digital and offline channels.
- **Digital Transformation:** Commended for pioneering initiatives like WhatsApp servicing and mobile app-based policy management.
- **Claims Efficiency:** Noted for high claim settlement ratios and rapid processing times, fostering trust and transparency.

Awards have been received at prestigious platforms such as the ET Insurance Summit, BFSI Awards, and FICCI Insurance Excellence Awards, underscoring Bajaj Allianz Life's commitment to innovation and customer satisfaction.

1.4 Conclusion

This report encapsulates the dual focus of the OJT assignment: leveraging data analytics to derive actionable insights and gaining a comprehensive understanding of Bajaj Allianz Life Insurance's operations. Through EDA and the Apriori algorithm, the project highlights patterns in customer behavior and proposes a recommendation model to enhance product personalization. Combined with insights into the company's structure, digital initiatives, and performance metrics, this work demonstrates the integration of analytical skills and industry knowledge to drive strategic decision-making and customer engagement.

1.5 Product Category Summary Tables

The following two tables summarize product categories offered by Bajaj Allianz Life Insurance, mapped by life stage and by product variant count.

Table 1: Customer Life Stage and Product Mapping

Customer Life Stage	**Product Type**	**Example Plan**
Young Professionals	Term Plan / ULIP	Smart Protect Goal, Goal Assure
Married with Dependents	Child Plan / Endowment	Young Assure, Save Assure
Pre-retirement (40–55)	Savings / Pension	Guaranteed Income Goal
Retired Individuals	Pension Plan / Annuity	Retire Rich
Rural / Low-Income	Micro Insurance / Simple Plans	POS Goal Suraksha

Table 2: Product Categories, Variants, and Examples

Product Category	**No. of Variants**	**Examples**
Term Plans	6	Smart Protect Goal, Life Secure
ULIPs	5	Goal Assure, Future Gain
Savings Plans	4	Save Assure, Guaranteed Income
Child Plans	2	Young Assure
Retirement Plans	3	Retire Rich
Health Riders	3	Critical Illness, ADB, Waiver

Chapter 2

Design and Development of an Insurance Recommendation System

2.1 Rationale for the Project

- **Complexity of Insurance Products:** The diverse range of insurance policy structures, including Term, Endowment, and Unit-Linked Insurance Plans (ULIPs), poses significant challenges for customers and agents in selecting optimal plans. A rule-based recommendation system streamlines this decision-making process, enhancing efficiency and accuracy.
- **Data-Driven Decision Making:** Insurance companies manage extensive datasets encompassing customer profiles and policy details. This project leverages the Apriori algorithm to transform raw data into actionable insights, facilitating informed product recommendations.
- **Limitations of Traditional Approaches:** Conventional methods, such as static brochures and manual consultations, often fail to address individual customer attributes, including age, income, gender, or geographic location. This system provides personalized recommendations based on these variables.
- **Enhanced Transparency and Explainability:** By employing a rule-based approach, the system delivers transparent recommendations, presenting users with explicit rules (e.g., *"Customers with a 10-year premium term frequently select a 20-year benefit term with 85% confidence"*), fostering trust and clarity.
- **Empowering Insurance Agents:** This tool equips agents from companies such as Bajaj Allianz, LIC, and HDFC with real-time, data-driven policy recommendations based on customer inputs, improving operational efficiency and customer satisfaction.

2.1.1 Data Generation and Exploratory Data Analysis (EDA)

Objectives

- Generate realistic synthetic life insurance data aligned with Bajaj Allianz Life's product portfolio.
- Conduct Exploratory Data Analysis (EDA) to examine customer profiles, product preferences, and policy performance.

- Identify patterns and correlations to support personalized product recommendations and customer segmentation.

Dataset Overview

The synthetic dataset comprises approximately **100,000 records** and **23 attributes**, encapsulating customer demographics, policy characteristics, financial details, and outcomes.

Key Variable Categories

Category	**Attributes**
Demographics	Age Group, Gender, Marital Status, Occupation, Education Level
Geographic	Location, Region
Financial	Income, Declared Income, Number of Dependents
Policy Details	Product, Policy Type, Policy Status, Benefit Term, Premium Term, Sum Assured, Mode of Premium Payment, Premium per Payment, Premium Term Category
Temporal	Date of Purchase (Years Ago), Time to Maturity
Outcomes	Claim History, Children

Table 2.1: Categorization of Attributes in the Synthetic Dataset

Attribute Notes

- **Product** and **Policy Type** align with Bajaj Allianz Life's offerings, including ULIPs and Term Plans.
- **Premium Term Category** classifies premium terms as Short, Medium, or Long.
- **Claim History** serves as a binary outcome variable for risk assessment and outcome modeling.

Data Generation Methodology

The synthetic dataset was generated using Python, employing a rule-based approach to ensure realistic and business-aligned customer-product relationships. The methodology incorporated:

- Mapping demographics and income levels to appropriate product types.
- Aligning policy terms with age groups and geographic regions.
- Correlating declared income with sum assured and premium amounts.

The implementation details are available in the source notebook `Bajaj_Life_Data_Generation_Notebook.ipynb`.

Tools Utilized

- Python libraries: `pandas`, `numpy`, `random`, `Faker` for synthetic data generation.

Example Code – Sum Assured Generation:

```
def generate_sum_assured(product):
    return {
        'Smart Protect Goal': np.random.randint(500000, 1500000),
        'Save Assure': np.random.randint(200000, 1000000),
        'Goal Assure': np.random.randint(400000, 1200000),
        'Child Advantage Plan': np.random.randint(300000, 1000000),
        'Income Assure': np.random.randint(250000, 800000),
        'Future Gain': np.random.randint(500000, 1200000)
    }[product]
```

Example Code – Premium Term Categorization:

```
if premium_term < 10:
    premium_term_cat = 'Short'
elif premium_term <= 20:
    premium_term_cat = 'Medium'
else:
    premium_term_cat = 'Long'
```

Generation Logic Examples

The following rules were applied to ensure realistic data generation:

Customer Attribute Logic	**Product Allocation**
Age 18–25, single, low income	Term Plans (e.g., Smart Protect Goal)
Age 26–35, middle income, married with children	ULIPs or Child Plans
Age 46–60, nearing retirement	Retirement or Guaranteed Income Plans
Declared income > 1000000₹, urban location	High Sum Assured ULIPs or Combination Products
Policy term calculation	`Time_to_Maturity = Benefit_Term - Date_of_Purchase`

Table 2.2: Rules for Synthetic Data Generation

Exploratory Data Analysis (EDA)

The EDA employed a range of visualization and statistical techniques to uncover insights:

- **Univariate Analysis:** Frequency distributions using count plots and histograms.
- **Bivariate Analysis:** Relationships such as Age vs. Product and Income vs. Sum Assured.

- **Multivariate Segmentation:** Facet grids and crosstabulations for multi-dimensional analysis.
- **Risk and Claims Analysis:** Examination of Claim History by Region and Occupation.
- **Advanced Segmentation:** Heatmaps analyzing Product, Age, Region, and Income interactions.

Example Python Code for Count Plots:

```
import seaborn as sns
import matplotlib.pyplot as plt

for col in categorical_columns:
    plt.figure(figsize=(8, 4))
    sns.countplot(data=df, x=col, order=df[col].value_counts().index,
    ↪ palette="Set2")
    plt.title(f"Distribution of {col}")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

Key Insights from EDA

The EDA revealed critical trends and patterns:

Insight Type	**Key Finding**
Demographics	Predominance of customers aged 36–45 with a balanced gender distribution
Popular Products	High prevalence of Smart Protect Goal (Term) and Goal Assure (ULIP)
Sum Assured Trends	Higher sum assured for incomes above 600000₹ and urban locations
Claims Patterns	Elevated claim history among business owners and ULIP policyholders
Policy Behavior	Younger customers favor shorter terms and flexible premium payment options

Table 2.3: Summary of Key EDA Insights

2.2 Recommendation Model Development

2.2.1 Objective

The objective of this project is to develop a sophisticated rule-based insurance recommendation system designed to recommend an optimal **Benefit Term** for life insurance policies, tailored to a customer's demographic and financial profile. This system aims to deliver personalized, data-driven recommendations to enhance decision-making for insurance advisors and digital platforms.

Throughout the internship, four distinct models were iteratively developed, progressing from manually crafted rule-based logic to advanced data-driven rule extraction utilizing the **Apriori algorithm** implemented in both Python and R environments.

Core Objectives

- Develop a robust recommendation engine for life insurance by identifying patterns in customer preferences and translating them into actionable, data-driven rules.
- Empower insurance advisors and digital platforms with tools to deliver personalized product recommendations based on comprehensive customer data inputs.
- Advance the application of statistical techniques, such as association rule mining, to address business challenges in policy design, customer segmentation, and decision support.

Project Goals

- Explore and refine rule-based recommendation methodologies, transitioning from static rule tables to dynamic, Apriori-driven rule extraction.
- Utilize both synthetic and realistic insurance datasets, incorporating key attributes such as Premium Term, Income, Gender, Occupation, and Marital Status.
- Implement the Apriori algorithm to identify frequent itemsets and derive association rules that reflect customer preferences for Benefit Terms.
- Develop intuitive, user-facing interfaces using `Streamlit`, `Tkinter`, and `R Shiny` to enable:
  - Input of customer profiles.
  - Delivery of personalized Benefit Term recommendations.
  - Transparent display of rule confidence and explanatory details.

The overarching aim is to simulate a scalable, data-driven recommendation system that supports insurance agents, customers, and digital platforms in making informed, transparent, and personalized product selections aligned with policyholder needs.

2.2.2 Model Evolution Overview

A recommendation system is a sophisticated algorithmic framework that leverages data patterns to suggest products, actions, or content to users. Such systems are integral to platforms like *Netflix*, *Amazon*, and *Spotify*. In this project, the concept is adapted to the insurance domain, where the system recommends an optimal **Benefit Term** (coverage duration) for life insurance customers based on their demographic and financial profiles.

Recommendation systems can be categorized as follows:

- **Content-Based Filtering:** Suggests items similar to those previously preferred by the user.

- **Collaborative Filtering:** Recommends based on the preferences of users with similar behavioral patterns.
- **Rule-Based Filtering (adopted in this project):** Utilizes logical patterns and conditional rules derived from data to drive recommendations.

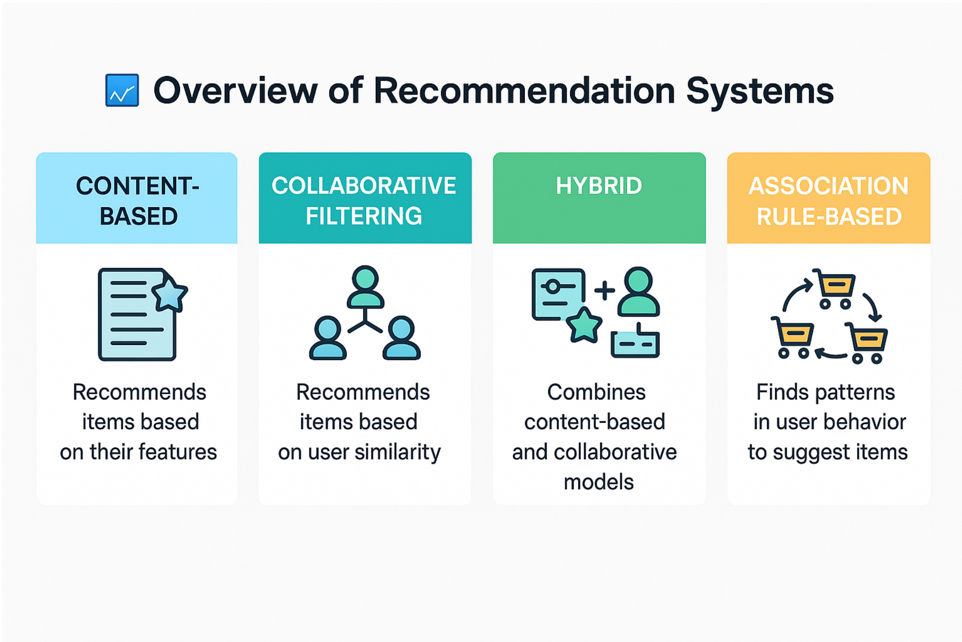


Figure 2.1: Conceptual Overview of the Recommendation System

This project employs a rule-based recommendation system powered by association rule mining, specifically tailored to recommend appropriate Benefit Terms for life insurance policies based on customer attributes.

2.2.3 Significance of the Apriori Approach in Insurance

The Apriori algorithm, a cornerstone of association rule mining, is employed to uncover relationships and co-occurrence patterns within large datasets. This approach is particularly effective in scenarios such as market basket analysis, where rules like *"If a customer buys bread and butter, they are likely to buy milk"* are derived. In this project, the same principle is applied to insurance data:

> *"If a customer has an income of 6–10L, is an engineer, and selects a premium term of 10 years, they are likely to choose a Benefit Term of 20 years."*

These rules are defined by:

- **Antecedents:** Input conditions (e.g., income, occupation).
- **Consequent:** Recommended Benefit Term.
- **Metrics:** Support, Confidence, and Lift to evaluate rule strength and reliability.

Advantages of the Apriori Approach

The Apriori algorithm is particularly well-suited for this project due to the following reasons:

- **Compatibility with Categorical Data:** Features such as Gender, Income, and Occupation are non-numeric, making them ideal for rule-based analysis.
- **Flexibility in Target Variables:** Unlike supervised learning methods, Apriori does not require a predefined target variable, allowing natural emergence of Benefit Term as the consequent.
- **Transparency and Interpretability:** Generated rules are straightforward and auditable, aligning with the regulatory requirements of the insurance industry.
- **Scalability:** Apriori efficiently processes large datasets by pruning infrequent patterns using minimum support thresholds.

Example of a Generated Rule

Rule	Confidence	Lift
If `premium_term=10`, `income=6-10L`, `occupation=engineer` ⇒ `benefit_term=20`	0.48	1.5

Table 2.4: Sample Association Rule Derived from the Apriori Algorithm

This rule indicates that 48% of customers matching the specified attributes select a 20-year Benefit Term, with a lift of 1.5, suggesting a strong and reliable recommendation.

Conclusion

The adoption of the Apriori algorithm facilitates an efficient, transparent, and data-driven approach to recommending insurance product features. By leveraging historical customer data, the system delivers personalized, explainable recommendations, enhancing decision-making for insurance advisors and digital platforms while aligning with industry standards for transparency and scalability.

2.2.4 Other Methods

Rationale for Selecting the Apriori Algorithm Over Alternative Models

While various machine learning and statistical approaches, such as logistic regression, decision trees, or collaborative filtering, could be applied to recommendation systems, the **Apriori algorithm** was chosen for this project due to its unique advantages in the context of insurance policy term recommendation. The key reasons are outlined below:

1. **Transparency and Interpretability:** The Apriori algorithm generates intuitive, human-readable association rules (e.g., "If Premium Term = 10 years and Income = 6–10L, then Benefit Term = 20 years") accompanied by clear metrics such as *support*, *confidence*, and *lift*. These metrics facilitate straightforward communication with business stakeholders and insurance agents, enhancing decision-making clarity.

2. **Unsupervised Pattern Discovery:** Unlike supervised models that require a predefined target variable, Apriori operates in an unsupervised manner, identifying co-occurrence patterns in customer insurance choices. This makes it particularly suitable for policy term recommendation, where labeled data may be limited or unavailable.

3. **Compatibility with Categorical Data:** Customer attributes in this domain, such as gender, income group, and marital status, are predominantly categorical. Apriori directly processes such data without requiring extensive preprocessing, unlike regression-based models that often necessitate encoding or transformation.

4. **Explainable Recommendations for Domain Experts:** In highly regulated sectors like insurance, explainability is paramount. Apriori's association rules provide transparent justifications for recommendations, such as "72% of customers with a Premium Term of 15 years selected a Benefit Term of 25 years." This fosters trust and compliance with regulatory requirements.

5. **No Dependency on Historical User Feedback:** Unlike collaborative filtering, which relies on prior episode-based data, Apriori does not require prior user-product interaction history. This eliminates the cold-start problem, making it effective for new or hypothetical customers.

Limitations of Multinomial Logistic Regression While multinomial logistic regression is a viable alternative, it presents several limitations in this context:

- **Assumption of Linear Relationships:** It assumes linear log-odds relationships between predictors and class probabilities, which may not adequately capture the complex, non-linear behaviors inherent in insurance decision-making.
- **Requirement for Labeled Data:** It necessitates labeled outputs (e.g., Benefit Term), which may not generalize well across diverse datasets with mixed categorical and numeric features.
- **Limited Interpretability:** The model produces coefficients and probabilities rather than intuitive "if–then" rules, making it less accessible for non-technical stakeholders such as insurance agents.

2.2.5 Models

As part of this internship, we developed four distinct versions of the insurance recommendation system. These models represent a progressive evolution—from static rule-based configurations to data-driven, algorithmically enhanced recommender systems implemented using both Python and R.

2.3 Application of Models

Industry Applications and Policy Implications

The proposed recommendation system offers substantial potential for adoption by insurance companies as well as policy and regulatory bodies. By integrating domain-specific rules with data mining techniques, the system supports enhanced decision-making, operational efficiency, and regulatory transparency.

Applications for Insurance Companies

- **Personalized Customer Engagement:** Tailors Benefit Term suggestions to individual customer profiles, leading to higher satisfaction and policy retention.
- **Agent Decision Support:** Empowers agents with real-time, data-driven recommendations during client interactions, whether in-person or over the phone.
- **Cross-Selling Enablement:** Identifies customer segments likely to purchase add-on plans such as ULIPs or extended benefit options.
- **Product Development Insights:** Provides actuaries and product teams with evidence-backed insights into potential new policy configurations.
- **Transparency and Compliance:** Rule-based logic enhances customer trust and supports auditability and regulatory compliance.

Applications for Policymakers and Regulators

- **Market Behavior Analysis:** Association rule insights facilitate understanding of consumer preferences across demographic segments, informing inclusive policy frameworks.
- **Regulatory Transparency:** Promotes responsible selling practices through explainable AI, reducing the risk of mis-selling.
- **Innovation and Sandboxing:** Offers a potential prototype for digital InsurTech tools under public-private partnership models and sandbox programs.
- **Program Impact Assessment:** Supports evaluation of government-backed insurance schemes like PMJJBY and Bima Sugam through data-derived adoption metrics.

This project demonstrates the integration of statistical logic, machine learning, and intuitive user interface design to create responsible, scalable recommendation systems aligned with industry goals and regulatory standards.

Model Evolution Summary

Model Version	Approach	Tech Stack	Dataset Source	Recommendation Logic
Model 1	Apriori on all valid combinations	Python + Tkinter	Synthetic combinations	Association Rule Mining
Model 2	Apriori on real-world-like data	Python + Streamlit	Semi-synthetic dataset	Real-time Rule Mining + Interactive GUI
Model 3	Rule-based logic implementation	R + Shiny	Synthetic dataset	Conditional Logic + Rule Integration

Table 2.5: Summary of Recommendation Model Versions

2.3.1 Model 1: Apriori-Based Recommender System (Tkinter GUI)

Model 1 marked the transition from manually coded logic to a data-driven recommendation system, leveraging the **Apriori algorithm** to extract actionable rules from a comprehensive synthetic dataset. This implementation featured a full-stack desktop application built using Python's Tkinter framework.

Dataset Design and Preparation

A synthetically generated dataset was created to simulate real-world customer diversity by combining key life insurance-related demographic variables. The included attributes were:

- Premium Term
- Gender
- Income Group
- Occupation
- Marital Status
- Location
- Number of Children

Each record was tagged with a corresponding Benefit Term based on plausible domain knowledge.

Methodology

The methodology was composed of two integrated components: backend rule mining and frontend user interaction.

A. Backend Rule Mining Using Apriori

- **Step 1: Synthetic Data Generation**
  Constructed a combinatorial dataset by programmatically generating all plausible combinations of 7 categorical attributes.
- **Step 2: Data Preprocessing**
  All categorical values were standardized and formatted for compatibility with the Apriori algorithm.
- **Step 3: Transaction Encoding**
  Employed `TransactionEncoder` from the `mlxtend.preprocessing` module to convert data into a binary transaction matrix.
- **Step 4: Frequent Itemset Mining**
  Used the `apriori()` function to discover itemsets with minimum support ≥ 0.005.
- **Step 5: Association Rule Extraction**
  Applied `association_rules()` with constraints:
  - Confidence ≥ 0.4
  - Lift > 1.0
  - Consequent must include a Benefit Term

  Final rules were stored in `filtered_rules.csv` for GUI integration.

B. Frontend Interface Using Tkinter

The graphical user interface was designed for intuitive use and deployed as a standalone desktop application.

- **User Input:** Dropdown menus for all profile features. A 'Submit' button initiates rule-based recommendation.
- **Matching Algorithm:** Inputs are converted to label format. Matching rules from `filtered_rules.cs` are retrieved based on full antecedent matches.
- **Recommendation Display:** Top three matched Benefit Terms are displayed with:
  - Confidence Score
  - Lift Value
  - Rule Explanation
- **Logging:** All recommendations are logged into `recommendation_log.csv` with timestamp and metadata.

Sample Rule Output

`If {'premium_10', 'income_6-10l', 'occupation_engineer'}` ⇒ `{'benefit_20'}`
Confidence: 0.48 **Lift:** 1.52
Interpretation: Customers with these characteristics have shown preference for a Benefit Term of 20.

Results and Observations

- Generated a wide set of interpretable rules from synthetic data covering a broad feature space.
- GUI effectively rendered personalized suggestions using rule-based explanations.
- Inclusion of both confidence and lift provided flexible evaluation options.
- Fully local execution ensured privacy and responsiveness.

Model Strengths and Limitations

Strengths:

- **Adaptive Rule Learning:** New combinations or business rules can be integrated by retraining, eliminating the need for manual code adjustments.
- **High Explainability:** Every recommendation is backed by transparent logic with measurable metrics.
- **Robust Coverage:** Synthetic data covers rare edge cases and supports multiple ranking preferences.
- **Traceable Recommendations:** Logging ensures full traceability and auditability of the decision process.

Limitations:

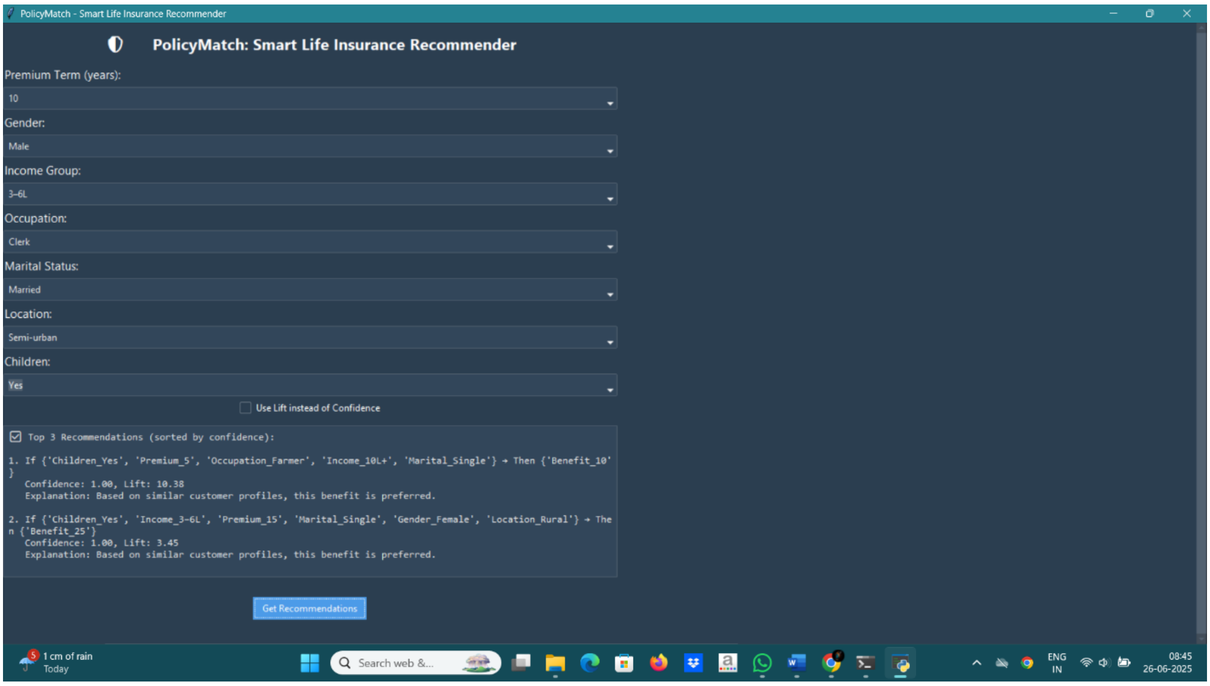


Figure 2.2: Model 1: Tkinter-Based Recommendation Interface

- Reliance on synthetic data reduces real-world predictive accuracy and generalizability.
- The desktop interface limits portability and real-time integration compared to web-based solutions.
- Lack of fallback mechanism in the absence of matching rules.
- Absence of continuous learning—requires full reprocessing for any dataset updates.

2.3.2 Model 2: Apriori-Based Recommender (Streamlit Interface)

The insurance industry offers a wide variety of policies with different premium and benefit terms. This version of the model was developed to deliver personalized benefit term recommendations using a clean and interactive **Streamlit web interface**.

- The model applies the **Apriori algorithm** to discover associations between user demographic/financial features and insurance `Benefit_Term`.
- A **Streamlit-based GUI** collects user input and presents relevant benefit term suggestions with rule strength indicators.

Dataset Description

The dataset used in this model is a synthetically generated life insurance dataset containing **151,200 customer records**. It simulates realistic combinations of user profiles and serves as the foundation for rule mining.

Dataset Overview:

- **File Name:** `large_insurance_dataset.csv`
- **Total Records:** 151,200
- **Format:** CSV file with 7 input features and 1 target feature

Features Used in Rule Mining

Feature Name	Type	Description
Premium_Term	Numeric	Number of years the customer pays premiums
Gender	Categorical	Male / Female
Income	Categorical	Income range: <3L, 3–6L, 6–10L, 10L+
Occupation	Categorical	e.g., Engineer, Teacher, Doctor, Business Owner
Marital_Status	Categorical	Single / Married
Location	Categorical	Urban / Semi-urban / Rural
Children	Categorical	Yes / No
Benefit_Term	Numeric	**Target variable:** Number of years of life coverage

Table 2.6: Feature Overview for Model 2 Dataset

Each feature was preprocessed and transformed into a standardized labeled string format (e.g., `income_6-10l`, `occupation_engineer`) for compatibility with the Apriori algorithm. These formatted inputs enabled efficient rule extraction and matching.

Methodology

This section outlines the overall methodology used to build a rule-based insurance benefit term recommendation system using the **Apriori algorithm**. The implementation was carried out in Python and deployed via a Streamlit web interface.

Step 1: Data Transformation and Cleaning

- All categorical values were converted to lowercase.
- Extra spaces were removed using `.str.strip()`.
- Each feature was prefixed to create consistent labels (e.g., `income_6-10l`, `gender_male`, `premium_10`).
- This formatting minimized rule duplication and ensured compatibility with Apriori inputs.

Step 2: (Optional) Exploratory Data Analysis (EDA)

- Conducted in a separate notebook to understand key patterns such as:
  - Most common Premium Terms
  - Benefit Term distribution across demographics
- Visualizations included histograms, count plots, and heatmaps created using Seaborn and Matplotlib.
- Helped validate patterns later extracted via Apriori.

Step 3: Boolean Encoding for Apriori

- Using `TransactionEncoder` from `mlxtend`, the dataset was converted into a transaction-style binary matrix.

- Each record became a list of feature labels like:

```
['gender_male', 'occupation_engineer', 'income_6-10l', 'benefit_20']
```

- All rows were encoded to produce a binary dataframe with shape: `151,200 rows × 7 features`.

Step 4: Frequent Itemset Generation (Apriori)

- Frequent itemsets were generated using:

  `apriori(df_encoded, min_support=0.005, use_colnames=True)`
- This step identified all customer attribute combinations that occurred frequently.

Step 5: Rule Extraction and Filtering

- Rules were extracted from itemsets using:

  `association_rules(frequent_itemsets, metric='confidence', min_threshold=0.4)`
- Only rules with `Benefit_Term` in the consequent were retained.
- Additional filters applied:
  - **Confidence** ≥ 0.4
  - **Lift** > 1.0

Sample Rule Output:

`If {premium_10, income_6-10l, occupation_engineer} → Then {benefit_20}`
Confidence: 0.48

Step 6: Recommendation Logic (Real-Time Matching via Streamlit)

- User inputs (e.g., Gender, Income, Occupation) are transformed into Apriori-compatible labels.
- System searches for rules where:
  - `Antecedent` ⊆ user input
  - Sorted by confidence (or lift) for prioritization
- Top Benefit Term(s) are recommended.

Example Recommendation And Interface Screnshot

`IF ['premium_10', 'income_6-10l', 'occupation_engineer']`
`THEN ['benefit_20']` **Confidence:** 0.48

Results & Limitation (Model 2)

- The model successfully extracted over 500 association rules from the full dataset.
- Most common recommendations were in the 20–30 year `Benefit Term` range.
- Higher-income professionals (e.g., doctors, engineers) were typically linked to longer benefit terms.
- The Streamlit interface allowed for real-time recommendations with explainable rule matches.
- Matching rules were ranked by confidence, ensuring transparency in decision-making.

Strengths of Model (Apriori + Streamlit):

- Based on a realistic synthetic dataset, improving relevance to business needs.
- Fully interpretable rule-based logic using the Apriori algorithm.
- Accessible through a user-friendly web interface built in Streamlit.
- Rule thresholds (support, confidence, lift) can be tuned to adjust sensitivity.
- Recommendations are aligned with real-world customer patterns and easy to justify.

Limitations of Model:

- Still reliant on synthetic data — does not reflect real-time customer behavior.
- May return no recommendation if no rule matches the input combination.
- Limited scope — only predicts `Benefit Term`, not full policy structure.
- Apriori algorithm can be memory- and computation-intensive for very large datasets.

2.3.3 Model 3: Apriori-Based Recommender (R Shiny)

This fourth version of the recommendation model was implemented using **R Shiny**, aiming to explore how association rule mining can be applied in a different programming ecosystem while maintaining model explainability and user interactivity.

Key Objectives:

- Replicate the Apriori-driven recommendation logic from the earlier Python models using `arules` in R.
- Build a fully functional, browser-accessible web application using `R Shiny`.
- Allow users to upload insurance datasets, perform EDA, and receive `Policy Term` recommendations based on mined association rules.
- Demonstrate cross-platform adaptability and enhance the model's reach to R-based analytics teams.

This model emphasizes both **explainability** and **usability**, enabling non-technical users (such as agents or underwriters) to interact with a rule-based, data-informed recommendation system.

Column Name	Description
`age`	Numerical customer age
`gender`	Gender: Male / Female
`declared_income`	Annual income in INR
`premium_term`	Number of years premium is paid
`policy_term`	Number of years the policy provides coverage
`booking_frequency`	Premium payment frequency: Monthly, Quarterly, etc.
`location`	Customer location: Urban / Semi-urban / Rural
`policy_type`	Product category (e.g., ULIP, Term Plan)
`current_status`	Marital status: Married, Unmarried, Widow

Table 2.7: Expected Input Columns for R Shiny App

Dataset Description

The application accepts insurance data in `.csv` format uploaded by the user. The dataset is expected to contain key attributes relevant to life insurance recommendations, such as:

Derived Features (Inside App)

To enable transactional rule mining, two key attributes are derived within the application:

- `age_group`: Created using `cut()` — e.g., 18–30, 31–40, etc.
- `income_group`: Generated using `case_when()` — grouped as 0–3L, 3–6L, etc.

These transformations allow the dataset to be converted into a transaction format suitable for association rule mining using the `arules` package in R.

R Shiny Interface Features

The R Shiny application consists of a well-structured and interactive interface designed to make the recommendation model user-friendly and accessible. The layout is divided into three key tabs, each with distinct functionalities:

Input Panel (Sidebar):

Users provide their inputs through dropdown menus located in the sidebar panel. The following attributes are required:

- `Premium Term` (in years)
- `Age Group` (e.g., 18–30, 31–40)
- `Income Group` (e.g., 0–3L, 3–6L, 6–10L)
- `Gender` (Male / Female)
- `Location` (Urban, Semi-Urban, Rural)
- `Current Marital Status` (Married, Unmarried, Widow)

- `Booking Frequency` (Monthly, Quarterly, etc.)

An `actionButton` triggers the rule-based recommendation engine once all inputs are selected.

Recommendations Tab:
This tab displays the real-time recommendation results after processing user inputs. Features include:

- A short summary message outlining the selected premium term and recommendation context
- A ranked table of top-matching rules sorted by `confidence × lift`, with:
  - Full rule explanation (e.g., `premium_term=10, income=6-10L ⇒ policy_term=25`)
  - Confidence score (probability of correctness)
  - Lift value (rule strength vs. random chance)
- If no exact match is found, a fallback recommendation is provided using the most common `policy_term` for the selected `premium_term`.

Data Summary Tab:
This tab supports exploratory data analysis with visual insights:

- Histograms for:
  - Age
  - Declared Premium
  - Premium Term
  - Gender
  - Booking Frequency
  - Location
  - Marital Status
- Each chart is followed by an auto-generated conclusion, such as:
  - "Most users belong to the 31–40 age group"
  - "Quarterly booking frequency is most popular"

These visual summaries help users explore the structure of the dataset before interpreting rule-based outcomes.

Interface Screenshot

Results & Limitation (Model 3)

- The app accurately returned top rule-based policy term suggestions for most valid input combinations.
- Built-in fallback logic ensured users never encountered empty results, even for rare profiles.

- Embedded EDA visualizations helped validate underlying data distributions and usage trends.
- Users were able to interpret the rationale behind each suggestion through clearly formatted rule tables.

Strengths of Model (R Shiny + Apriori):

- Fully implemented in R, allowing broader access for teams that work within the R ecosystem.
- Real-time rule filtering with dynamic confidence and lift-based sorting.
- Built-in fallback mechanism improves robustness in edge cases.
- Embedded EDA charts provide additional insights before rule interpretation.
- Tab-based UI layout improves usability and content separation.

Limitations of Model 3:

- Requires users to upload a CSV file; no native data simulation or storage.
- Fewer deployment options for scalable cloud/web use compared to frameworks like Streamlit.
- Rule logic must be re-generated (re-running Apriori) when new datasets are uploaded.
- Slightly more complex setup required for users unfamiliar with the R environment.

2.3.4 Comparative Insights Across All Models

To evaluate the progression and effectiveness of the four recommendation models developed during the internship, the following comparative analysis summarizes their key aspects, capabilities, and trade-offs.

Key Comparative Insights

- **Model 1** introduced Apriori mining, demonstrating that rules can be learned from data, but lacked real-world grounding.
- **Model 2** offered the best balance of realistic data, explainability, and interactive interface—making it suitable for practical business use.
- **Model 3** proved that the same Apriori logic could be extended to R-based ecosystems, adding fallback handling and built-in EDA capabilities.

Comparison Table

2.3.5 Overall Results and Observations

- **General Observations:**
  - Across all models, the most commonly recommended `Benefit Terms` or `Policy Terms` ranged between 20 to 30 years—particularly for younger, middle-income groups.

Feature	**Model 1** (Apriori + Tkinter)	**Model 2** (Apriori + Streamlit)	**Model 3** (Apriori + R Shiny)
Data Source	Synthetic (combinations)	Uploaded CSV	Uploaded CSV
Algorithm Used	Apriori (Python)	Apriori (Python)	Apriori (R)
UI Framework	Tkinter	Streamlit	R Shiny
Rule Discovery	Automated	Automated	Automated
Target Variable	Benefit Term	Benefit Term	Policy Term
Explainable Logic	Rule-based	Rule-based	Rule-based
Cold Start Handling	None	None	Fallback Rule
User Interaction	Moderate	Interactive Web UI	Rich Tabbed UI
EDA Support	No	Yes	Yes
Best Use Case	Rule Mining Concept	Realistic Application	R-based Analyst Use

Table 2.8: Comparative Analysis Across All three Models

  - Higher-income professionals (e.g., engineers, doctors) were more frequently associated with longer coverage durations.
  - Shorter premium terms (e.g., 10–15 years) were often linked to longer benefit durations, indicating user preference for limited payment periods with extended protection.

- **Model-Specific Output Trends:**
  - **Model 1 (Apriori + Tkinter):** Generated over 500 frequent itemsets from synthetic combinations, automating logic but sometimes producing unrealistic or redundant rules.
  - **Model 2 (Apriori + Streamlit):** Based on a more realistic synthetic dataset, returned filtered rules with high confidence. Streamlit enabled real-time, user-friendly interaction.
  - **Model 3 (Apriori + R Shiny):** Delivered accurate recommendations with fallback logic. Tab-based UI and built-in EDA enhanced both usability and analytical interpretation.

- **Practical Takeaways:**
  - Models using the Apriori algorithm outperformed manual rule logic by offering explainable, data-driven patterns.
  - `Confidence` and `Lift` proved effective in ranking and interpreting recommendations.
  - Web-based frameworks like `Streamlit` and `R Shiny` greatly improved accessibility and engagement.
  - The introduction of fallback logic in Model 4 enhanced model robustness—ensuring outputs even for edge-case inputs.

2.3.6 Overall Limitations

While the project successfully demonstrated multiple approaches to building a rule-based insurance recommendation system, several overarching limitations were identified across all models:

1. Use of Synthetic Data

- All models were developed using synthetic datasets.
- Although realistic patterns and customer logic were simulated, real-world factors like behavioral variance, economic shifts, and product constraints could not be fully replicated.

2. Cold Start Problem

- For rare or unusual customer profiles, the Apriori-based models often returned no recommendation due to lack of matching rules.
- Only Model 4 attempted to mitigate this with a fallback recommendation.

3. Static Rule Framework

- Rules generated by Apriori remain static unless re-trained with new data.
- This is a limitation in dynamic markets where product lines and customer behavior evolve frequently.

4. Limited Output Scope

- The models focused primarily on predicting `Benefit Term` or `Policy Term`.
- A complete recommendation system should also predict:
  - Policy Type
  - Sum Assured
  - Riders or Add-ons
  - Premium Payment Frequency

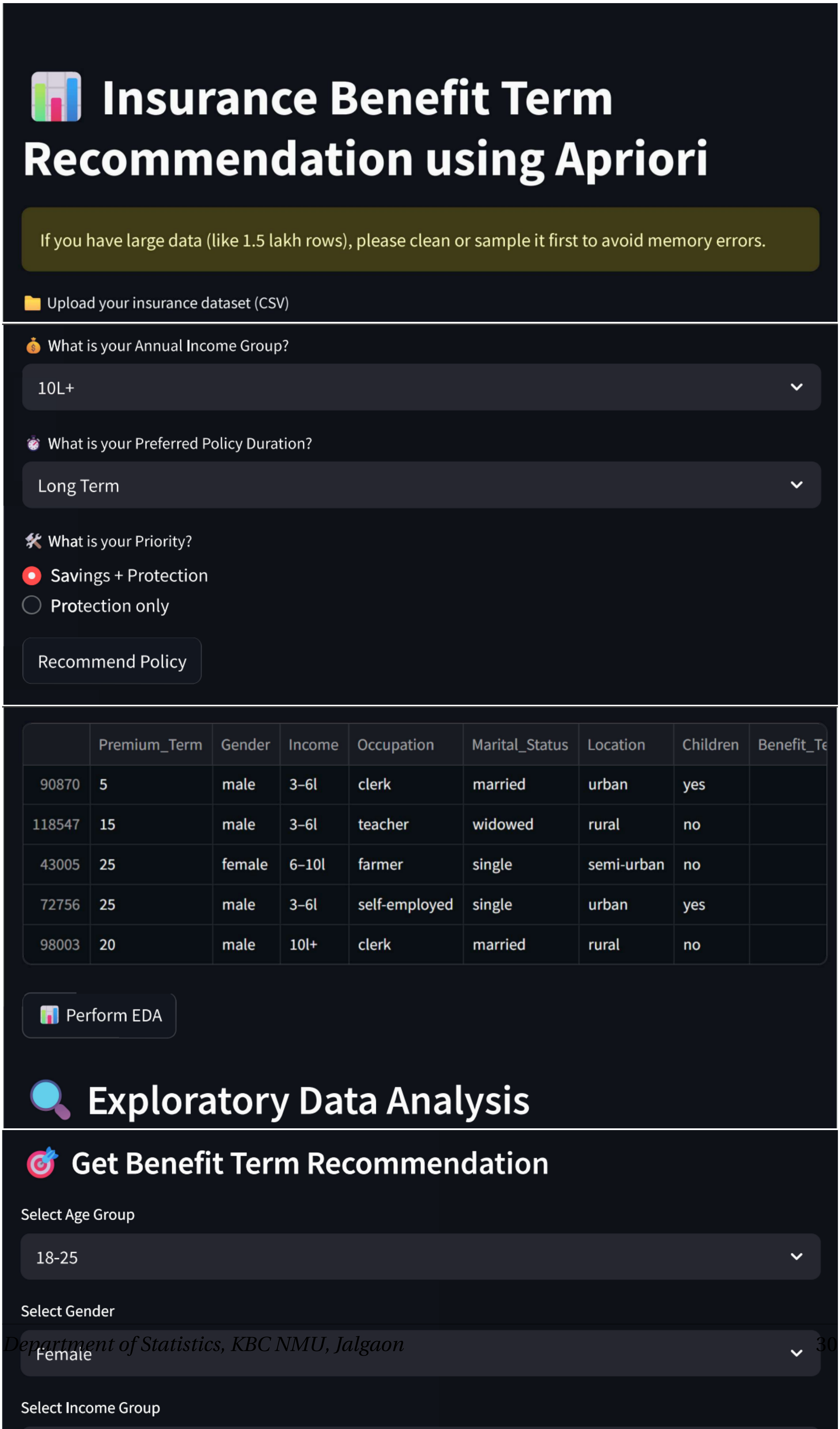
5. Performance & Scalability

- For large datasets (especially Models 2 and 3), the Apriori algorithm became memory- and computation-intensive.
- Without rule pruning or dimensionality reduction, real-time deployment becomes challenging.

6. No Learning from Feedback

- The current system does not learn or improve from user interactions.
- It lacks reinforcement learning, feedback loops, or adaptive personalization capabilities that would improve accuracy over time.

These limitations point to key areas for future enhancement, particularly for deploying the system in real-world insurance environments.



	Premium_Term	Gender	Income	Occupation	Marital_Status	Location	Children	Benefit_Te
90870	5	male	3–6l	clerk	married	urban	yes	
118547	15	male	3–6l	teacher	widowed	rural	no	
43005	25	female	6–10l	farmer	single	semi-urban	no	
72756	25	male	3–6l	self-employed	single	urban	yes	
98003	20	male	10l+	clerk	married	rural	no	

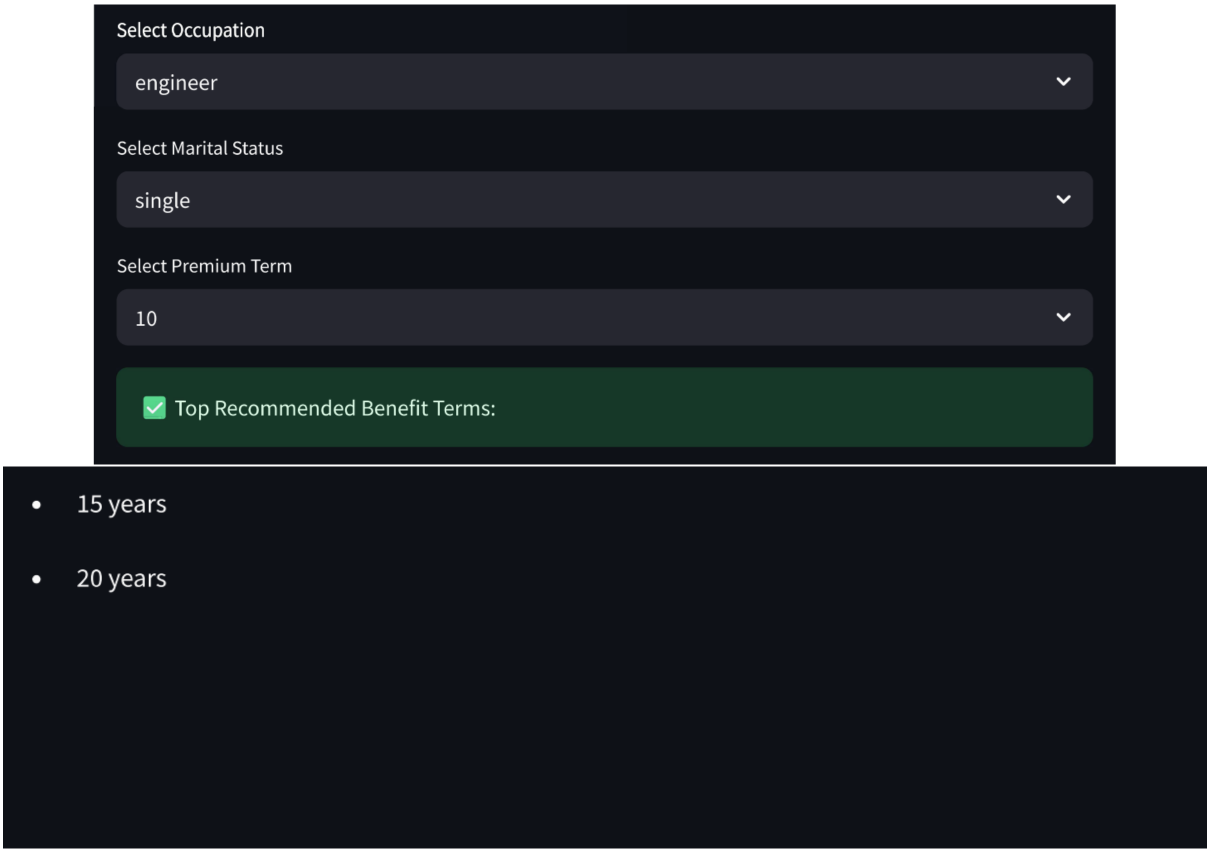
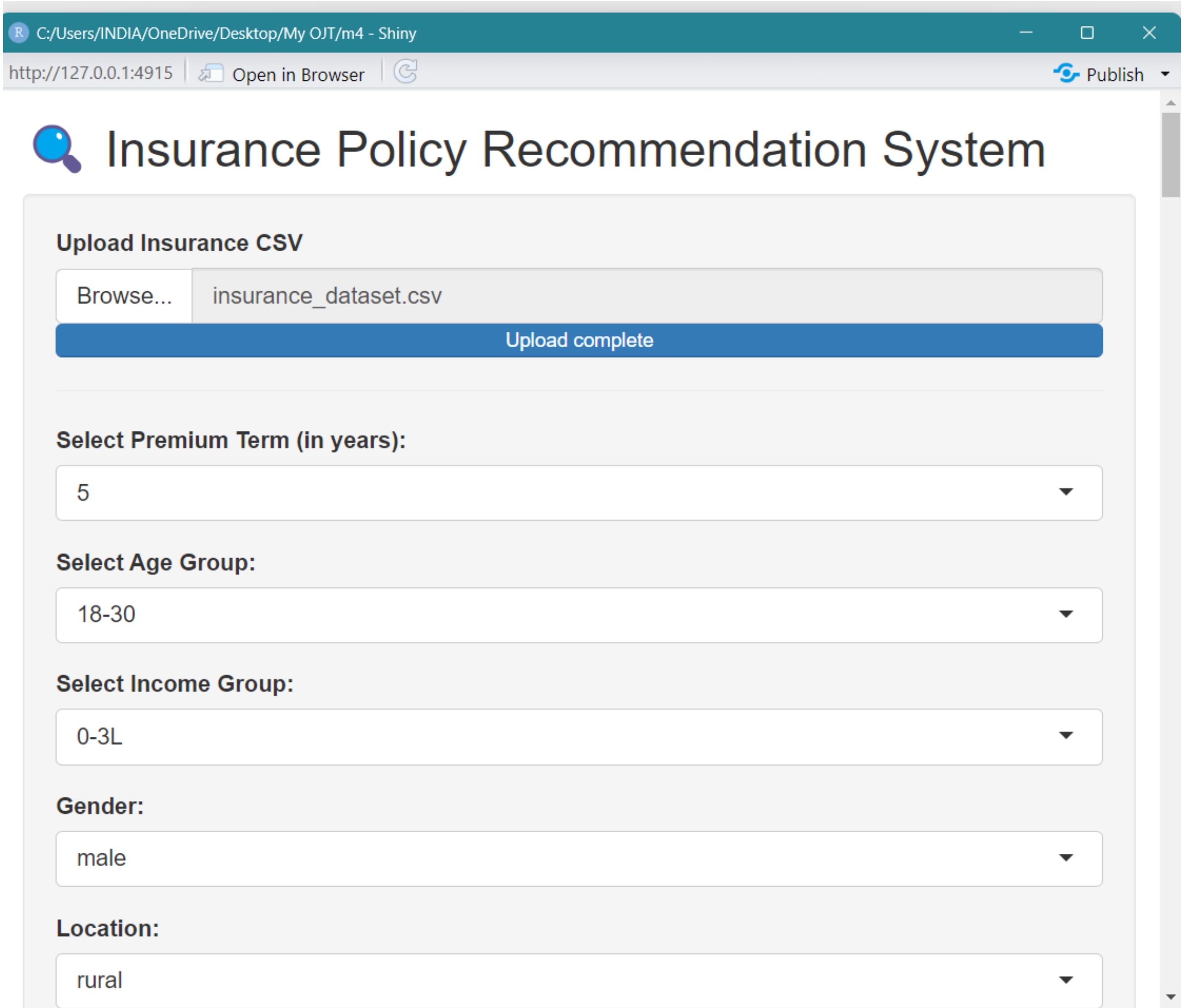
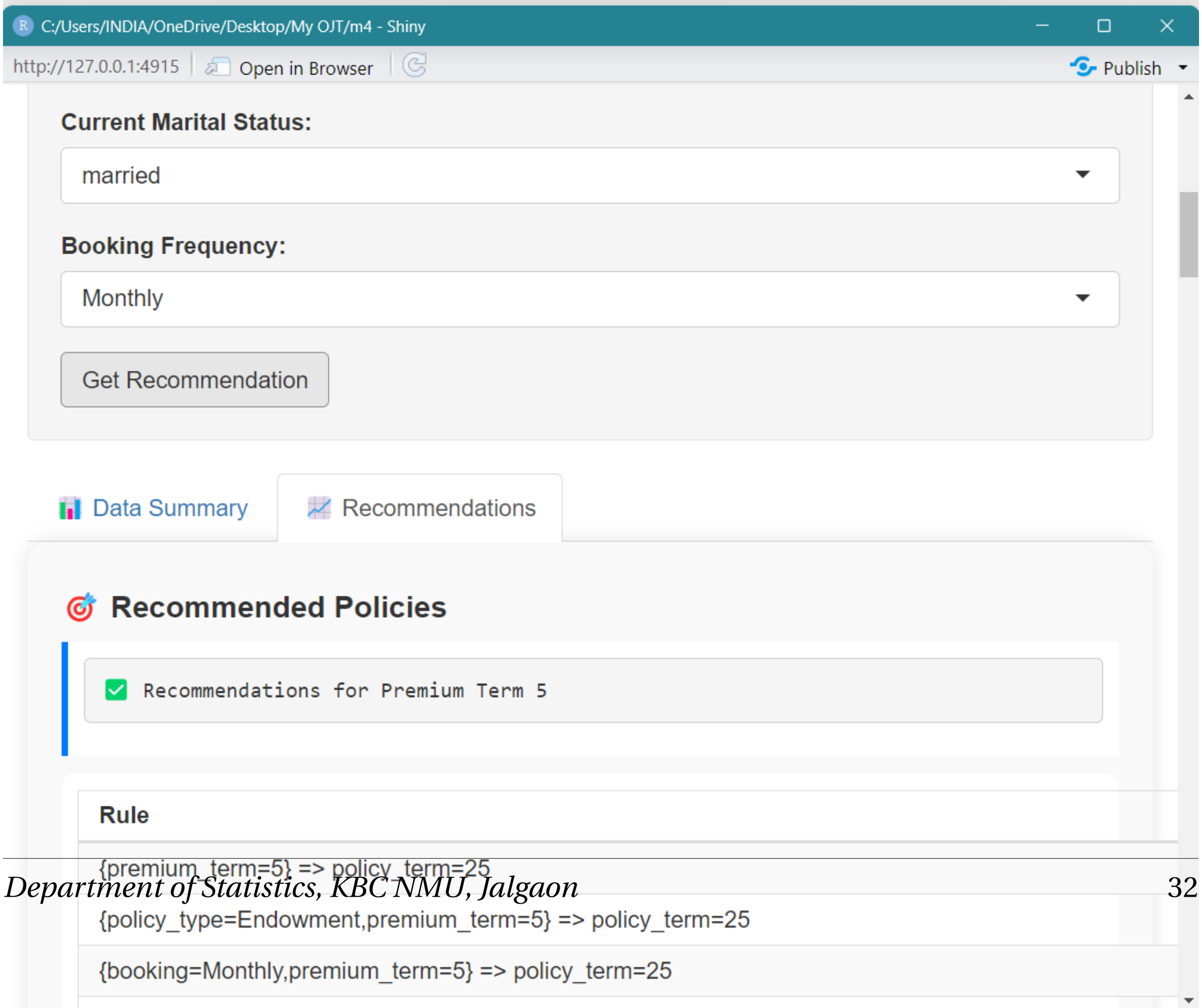


Figure 2.3: Streamlit Interface: User selects profile and receives Benefit Term recommendation


C:/Users/INDIA/OneDrive/Desktop/My OJT/m4 - Shiny
http://127.0.0.1:4915
Open in Browser
Publish
Insurance Policy Recommendation System
Upload Insurance CSV
Browse...
insurance_dataset.csv
Upload complete
Select Premium Term (in years):
5
Select Age Group:
18-30
Select Income Group:
0-3L
Gender:
male
Location:
rural


C:/Users/INDIA/OneDrive/Desktop/My OJT/m4 - Shiny
http://127.0.0.1:4915
Open in Browser
Publish
Current Marital Status:
married
Booking Frequency:
Monthly
Get Recommendation
Data Summary
Recommendations
Recommended Policies
Recommendations for Premium Term 5
Rule
{premium_term=5} => policy_term=25
{policy_type=Endowment,premium_term=5} => policy_term=25
{booking=Monthly,premium_term=5} => policy_term=25

Chapter 3

Challenges Encountered During Internship and Model Development

3.1 Overview of Obstacles

The development of a rule-based insurance recommendation system during the internship at Bajaj Allianz Life Insurance presented a range of technical and conceptual challenges. These spanned data acquisition, algorithmic scalability, model implementation, and user interface optimization. Each obstacle was addressed systematically, contributing to the robustness of the final system and fostering significant professional growth.
The challenges provided critical learning opportunities, enhancing the system's technical resilience and adaptability while aligning with industry standards.

1. **Limited Access to Real Insurance Data**
   - *Challenge:* Due to strict confidentiality policies, real insurance datasets from LIC or SBI Life were unavailable despite formal requests.
   - *Solution:* Developed synthetic datasets using Python libraries (`faker`, `random`), informed by product brochures and official websites to ensure domain relevance.
2. **Imbalanced Data Distributions**
   - *Challenge:* Initial synthetic datasets exhibited unrealistic distributions for features such as age, income, and dependents, limiting model accuracy.
   - *Solution:* Refined simulation logic to mirror real-world insurance product preferences:
     - Younger customers prioritized term plans.
     - Parents favored child policies.
     - Higher-income individuals leaned toward ULIPs.
3. **Memory Constraints with Apriori Algorithm**
   - *Challenge:* Large datasets (150,000+ records, 7+ features) caused `MemoryError` during rule mining with the Apriori algorithm.
   - *Solution:*
     - Reduced dataset size through sampling (e.g., 25,000 records).
     - Minimized encoded variables to optimize memory usage.

  - Dynamically adjusted `min_support` and confidence thresholds.

4. **Sparse Rule Generation for Edge Cases**
   - *Challenge:* Certain user profiles failed to generate recommendations due to sparse rule outputs.
   - *Solution:*
     - Enhanced synthetic data variability to cover edge cases.
     - Lowered `min_support` to capture less frequent patterns.
     - Introduced fallback logic in Model 4 for default recommendations.
     - Incorporated exception handling in the user interface.

5. **Validation of User-Uploaded Datasets**
   - *Challenge:* User-uploaded datasets often lacked essential columns (e.g., `premium_term`, `income`, `policy_term`).
   - *Solution:* Implemented robust validation logic in Streamlit and R Shiny applications:
     - Verified presence of required fields.
     - Provided clear, user-friendly error messages.
     - Prevented rule generation until validation was successful.

6. **Performance Bottlenecks in Streamlit**
   - *Challenge:* Real-time operations (e.g., encoding, rule mining) in Streamlit caused slow performance or application crashes.
   - *Solution:*
     - Restricted dataset sizes for processing.
     - Optimized filtering and preprocessing logic.
     - Streamlined rule comparison algorithms.

7. **Navigating Unfamiliar Concepts**
   - *Challenge:* Association rule mining and recommendation systems were initially unfamiliar, with early errors (e.g., misspelling "Apriori") slowing progress.
   - *Solution:* Through rigorous study, practice, and mentorship:
     - Mastered Apriori algorithm workflows.
     - Clarified distinctions between LHS (antecedent) and RHS (consequent).
     - Developed proficiency in metrics such as support, confidence, and lift.

3.2 Internship Experience and Key Learnings

The one-month On-the-Job Training (OJT) at **Bajaj Allianz Life Insurance** provided a transformative opportunity to design and deploy an end-to-end insurance recommendation system. This project bridged theoretical knowledge with practical application, integrating concepts from statistics, actuarial science, business intelligence, and data science using Python and R.

1. **Technical Proficiencies Developed**

- Generated realistic synthetic datasets in Python to emulate life insurance customer profiles and product portfolios.
- Designed rule-based logic using decision tables and `if-else` structures for initial models.
- Implemented the Apriori algorithm in Python (`mlxtend`) and R (`arules`) for association rule mining.
- Developed interactive user interfaces with `Streamlit` (Python) and `Shiny` (R) for real-time recommendation delivery.
- Mitigated scalability challenges through:
  - Dataset sampling (e.g., 25,000 records from 150,000).
  - Threshold optimization for support and confidence.
  - Efficient transaction encoding and filtering techniques.

2. **Conceptual Insights Gained**

- Deepened understanding of life insurance product structures:
  - Term Insurance vs. Endowment Plans.
  - Premium and benefit term relationships.
  - Riders, booking frequency, and policy term dynamics.
- Analyzed the influence of demographic factors (age, gender, income, marital status) on insurance purchasing patterns.
- Mastered data mining concepts, including support, confidence, and lift, for evaluating rule-based models.

3. **Professional and Analytical Development**

- Enhanced skills in pattern recognition and data-driven decision-making.
- Translated complex rules into actionable business insights for insurance recommendations.
- Improved technical documentation and professional report-writing capabilities.
- Fostered collaborative problem-solving through teamwork with diverse peers.

4. **Holistic Internship Impact**

- Applied statistical and actuarial principles in a real-world business context.
- Explored practical applications of recommendation systems in the insurance industry.
- Delivered functional solutions by integrating business logic with advanced technical tools.

This internship cultivated technical expertise and a solution-oriented mindset, equipping me to address complex challenges in insurance analytics, actuarial modeling, and data-driven decision-making.

3.3 Summary of Achievements

The internship at **Bajaj Allianz Life Insurance** provided a platform to apply advanced statistical techniques and data science methodologies in a professional insurance context. The primary goal of developing a robust rule-based insurance recommendation engine was successfully achieved through a structured, multi-model approach.

- Evaluated multiple recommendation strategies, from manual rule-based systems to Apriori-driven models.
- Delivered functional applications using Python (`Streamlit`, `Tkinter`) and R (`Shiny`).
- Overcame technical challenges, including memory constraints, lack of real data, and edge-case handling.

Key Deliverables:

- A domain-informed synthetic insurance dataset.
- Four distinct recommendation models, evaluated for performance and trade-offs.
- Interactive graphical user interfaces for real-time insurance recommendations.

This project significantly advanced our expertise in association rule mining, exploratory data analysis, and software development, while fostering the ability to deliver practical, industry-relevant solutions.
The OJT experience marked a pivotal milestone in our professional journey, equipping me with the skills and confidence to excel as an industry-ready analyst in insurance analytics and data science.

3.4 Project Artifacts and Repository

All project components, including datasets, models, code, visualizations, and documentation, have been systematically organized into a comprehensive repository. The deliverables are accessible in the following formats:

- **Local Archive:** `OJT.zip` (shared offline or via secure file transfer).
- **GitHub Repository:** `https://github.com/RajshriDPatil/OJT_Bajaj_Allianz_Life`

Repository Structure
File Formats Included

- `.csv` – Synthetic datasets.
- `.ipynb`, `.py`, `.R` – Jupyter notebooks, Python scripts, and R scripts.
- `.docx`, `.pdf` – Internship report and reference materials.
- `.png` – Screenshots and user interface visualizations.

Submission Overview

- **Zipped Archive:** `OJT.zip`.
- **GitHub Repository:** `https://github.com/RajshriDPatil/OJT_Bajaj_Allianz_Life`.

Folder Name	Description
1. Data & EDA	Synthetic insurance datasets, preprocessing scripts, and exploratory data analysis notebooks (.ipynb).
3. Model 1	Apriori-based recommendation model with Tkinter GUI (tkinter_policy_recommender.py).
4. Model 2	Apriori-based recommendation using Streamlit application (apriori_update.py).
5. Model 3	R Shiny application (app.R) with fallback logic and histogram-based EDA.
report/	Final internship report (My_OJT_Report.docx).
references/	Academic research papers and notes (ref1.pdf, ref2.pdf).

Table 3.1: Structure of Internship Deliverables Repository

- **README File:** Included in both formats, detailing:
  - Project objectives and scope.
  - Descriptions of models and folder structure.
  - Instructions for running each component.
  - List of tools and technologies utilized.

Resources and Citations

The development of the insurance recommendation system relied on a combination of academic resources, technical tools, and domain expertise. The following references were critical to the project's success:

A. Python Libraries and Tools

- **NumPy, Pandas** – Numerical computation and data preprocessing.
  https://numpy.org, https://pandas.pydata.org
- **MLxtend** – Apriori algorithm for frequent itemset mining.
  https://rasbt.github.io/mlxtend/
- **Matplotlib, Seaborn** – Data visualization frameworks.
  https://matplotlib.org, https://seaborn.pydata.org
- **Streamlit** – Web application framework for Python-based interfaces.
  https://streamlit.io

B. R Libraries

- **Shiny** – Interactive web application development in R.
  https://shiny.rstudio.com
- **arules** – Association rule mining with Apriori in R.
  https://cran.r-project.org/web/packages/arules

C. Domain and Academic References

- **Bajaj Allianz Life Insurance** – Product brochures and policy structures.
  `https://www.bajajallianzlife.com`
- **LIC India** – Reference for synthetic dataset modeling.
  `https://www.licindia.in`
- **IRDAI Reports** – Regulatory insights and claim settlement benchmarks.
- **Association Rule Mining Concepts** – Support, confidence, and lift metrics.
  Article: GeeksforGeeks – Association Rules.
- **[Ref 1]** – Bhuvaneshwari, B., Deepthi, G. Y., & Keerthiana, K. (2023). Insurance recommendation system. International Journal of Innovative Science and Research Technology, 8(3), 1922-1924. https://www.ijisrt.com
- **[Ref 2]** – Parasrampuria, R., Ghosh, A., Dutta, S., & Sarkar, D. (2021). Recommending Insurance products by using Users' Sentiments. arXiv preprint arXiv:2108.06210.

D. AI-Assisted Research

- **ChatGPT (OpenAI)** – Supported research synthesis, debugging, exploratory data analysis, and LaTeX report structuring.